INTRODUCTION TO MACHINE LEARNING

Sample Course Notes for EduPath AI — PDF upload, chunking, embeddings and retrieval practice

Learning objectives

After studying these notes, a learner should be able to:

- Define machine learning and distinguish it from traditional rule-based programming.
- Explain supervised, unsupervised and reinforcement learning.
- Describe the basic steps in a machine learning workflow.
- Identify common evaluation measures for classification models.
- Explain overfitting and list practical ways to reduce it.

1. What is Machine Learning?

Machine learning (ML) is a branch of artificial intelligence in which computers learn patterns from data and use those patterns to make predictions or decisions. Instead of writing a separate rule for every possible case, a developer provides examples and chooses an algorithm that can learn from them.

Example: To identify spam email, a model can be trained using messages labelled “spam” or “not spam”. It learns patterns such as suspicious phrases, unusual links and sender behaviour. When a new message arrives, the model estimates which class it belongs to.

Traditional programming: data + rules → output. **Machine learning:** training data + learning algorithm → trained model; new data + trained model → prediction.

2. Main Types of Machine Learning

2.1 Supervised Learning

Supervised learning uses labelled examples. Each training example contains input data and the correct target answer. The model learns a relationship between inputs and targets so it can predict targets for unseen inputs.

Classification predicts a category, such as whether an email is spam. **Regression** predicts a numeric value, such as the expected price of a house.

2.2 Unsupervised Learning

Unsupervised learning works with data that does not have target labels. The algorithm looks for structure or patterns. Clustering is a common task: a system may group customers with similar purchasing behaviour. Dimensionality reduction is another task; it represents data using fewer variables while trying to preserve useful information.

2.3 Reinforcement Learning

Reinforcement learning trains an agent to make decisions by interacting with an environment. The agent takes an action, receives feedback in the form of a reward or penalty, and learns a strategy that aims to maximize long-term reward. It is used in areas such as game-playing research and robot control.

3. A Typical Machine Learning Workflow

Define the problem: Decide what needs to be predicted or grouped and how success will be measured.

Collect data: Gather relevant examples from reliable sources and check whether the data represents the intended task.

Clean and prepare data: Handle missing values, remove duplicates where appropriate, encode categories and scale features when needed.

Split the dataset: Use training data to fit the model and validation or test data to evaluate how well it generalizes.

Choose and train a model: Select an algorithm suitable for the task and learn its parameters from training data.

Evaluate the model: Compare predictions with known answers using metrics appropriate to the problem.

Improve and monitor: Adjust features or model settings, then monitor performance when the model is used with new data.

4. Classification Evaluation

A confusion matrix compares predicted classes with actual classes. For a binary classifier, it contains four outcomes:

Outcome	Meaning
True positive (TP)	The model predicts positive and the actual class is positive.
True negative (TN)	The model predicts negative and the actual class is negative.
False positive (FP)	The model predicts positive but the actual class is negative.
False negative (FN)	The model predicts negative but the actual class is positive.

Accuracy is the fraction of all predictions that are correct: (TP + TN) / (TP + TN + FP + FN). **Precision** is TP / (TP + FP), and **recall** is TP / (TP + FN). The F1-score is the harmonic mean of precision and recall. Accuracy alone can be misleading when one class is much more common than another.

5. Overfitting and Underfitting

Overfitting occurs when a model learns the training examples too closely, including noise or accidental details. It performs well on training data but poorly on new data. Possible remedies include using more representative training data, simplifying the model, regularization, cross-validation and early stopping.

Underfitting occurs when a model is too simple or has not learned enough from the data. It performs poorly on both training and unseen data. A more suitable model, better features or additional training may help.

> A good model should learn meaningful patterns that generalize to examples it has not seen before.

6. Short Review Questions

1. What is machine learning?
2. How does supervised learning differ from unsupervised learning?
3. Give one example each of classification and regression.
4. What is the purpose of splitting data into training and test sets?
5. Define a false positive.
6. Why can accuracy be misleading on an imbalanced dataset?
7. What is overfitting, and name two ways to reduce it.
8. What does an agent learn in reinforcement learning?

7. Key Terms

- **Algorithm:** a procedure used to learn patterns or produce an output.
- **Feature:** an input variable used by a model.
- **Label:** the target answer associated with a supervised training example.
- **Model:** the learned representation used to make predictions.
- **Training:** the process of fitting a model using data.
- **Generalization:** the ability to perform well on new, unseen examples.
- **Embedding:** a numerical representation of information, often used to compare semantic similarity.

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